

FINAL TERM Exam Questions

BIO301 Current SPRING 2021

MCQs from Past Papers

FaceBook
Whatsapp
Instagram
Youtube
@BiotechBrainybunch

1- DNA and cDNA
2- Molecular markers application.
3- Addition mutation and effects
4- Souther, northern and Western blots.
5- Long and short hair...punnet square
6- BFEL and COAS
7- DNA and cDNA
8- PCR
9- Mutation ki types
10- Bht repeat ho rhy achy sa kr Lyn wo
11- Major3 types of Mutation *long*
12- Function of STR *short*

13- Types of polymers DNA sequence *long*
14- Structure chamge in chromosome? *3 marks*
15- How Mutation alert Gene function *3marks*
16- Complications of inheritance pattern.
17- Co dominance.
18- PCR principle and reagents.
19- Autosomal recessive disease.
20- Difference bw DNA n cDNA (2
21- Function of drawing Pedigree (2
22- precautions to deal with RNA(3
23- principle and reagent of PCR(5
24- Nomemclature of Mutation (5
25- 3 safety rules of bio lab 2
26- Name two autosomal recessive disorder 3
27- Southern,western and northern blotting difference 3
28- Pcr principle 5
29- Mutation of genes and predicting that don't do any job 5
30- autosomal inheritence
31- hetrozygous dominant trait
32- penetrance
33- pliotropy and expressivity.
34- what is frequency range of ressessive dominant disorder (2marks)
35- What are the uses of PCR (3 marks)
36- Define linked and casuative single nucleotide polymorphism (3 marks)
37- What are the principle of PCR (5marks)
38- Give the applications of short range markers (5marks)
39- Define Cell lysis. 2
40- Difference between dna and cdna. 2
41- Aflp and rapd. 3
42- Southern , northern , western blotting. 3
43- Pcr principle.5

Facebook Whatsapp Instagram Youtube @Biotechbrainybunch

Email; mehwishshabirhussain07@gmail.com |

44- Mutation nomenclature rule according to nucleotide sequence , translation initiation codon , translation stop codon. 5

45- What is the intronic sequence with reference to exon (2)

46- If both parents plants are homozygous and one have green color seed and other have yellow color seed, then why the f1 generation have all yellow colored seeds. Justify. (2)

47-

48- What are inversions mutation? Write its effect. (3)

49- What is a heterozygous dominant trait? Give example (3)

50- How should you write mutations at different levels (DNA, RNA, proteins) according to HGNC rules? (5)

51- If **hh** long hair is recessive to **HH** short hair then, what would be the genotype of f1 generation? Draw punnet square. (5)

52- Pcr principle and reagents 5mark

53- Reagnts and their ratio in PCI 2marks

54- Addition mutation and effects 3marks

55- Short range uses of molecular markers 5marks

56- Use of edta and mgcl2

57- Define addition mutation and also describes it's effects?

58- What is heterozygous dominant trait also give an example?

59- Punnet square of a cross between homozygous short hairs H and long hairs h trait?

60- Difference between inorganic and organic extraction of DNA and also describes the steps of organic extraction of DNA?

FaceBook Whatsapp Instagram Youtube
@Biotechbrainybunch

Important Chemicals and their Functions

1. EDTA = binds with metal ions and prevent blood clotting
2. MgCl2 = component of lysis buffer and helps in lysis of cell membrane
3. SDS = inhibit nuclease activity and helps in cell lysis

Email; mehwishshabirhussain07@gmail.com |

4. PCI = maintains pH of solution and phase separation
5. Isopropanol = precipitates DNA
6. Proteinase K = digests proteins
7. DNase-free water = DNA degradation
8. TE buffer = dissolves DNA and inhibits nucleiotides degradation
9. Ethanol = used for washing pellet
10. Low TE Buffer = dilute DNA pellete
11. Trizol reagent = extraction of DNA, RNA & Proteins
12. Agaros = used for separation of large molecules e.g DNA
13. Bromophenol blue = gel loading dye
14. Ethidium Bromide = used as running buffer and visualizes DNA pattern

Important Diseases

Mitochondrial Diseases

1. Myoclonic Epilepsy

2. mitochondrial recessive ataxia

3. Leber heredity optic neuropathy

X-Linked Dominant diseases

1.Rett syndrome

2. Goltz syndrome

3. Alport syndrome

4. Fragile X-syndrome

Autosomal Dominant Diseases

Email; mehwishshabirhussain07@gmail.com |

1. Marfan Syndrome

2. Neurofibromatosis type I

X-Linked recessive Diseases

1. Heamophilia (commonly in men)

Important Lab techniques Reagents

Methods of DNA purification

1. Organic (Phenol-Chloroform) Extraction
2. Non-organic (Proteinase K and Salting out)
3. Chelex (Ion Exchange Resin) Extraction.
4. FTA Paper (Collection, Storage, and Isolation)
5. Silica Based (Silica exchange resin- Qiagen)

Face Book
Whatsapp
Instagram
Youtube
@Biotechbrainybunch

Reagents for DNA Extraction (Organic Method)

1. Lysis Buffer
2. EDTA
3. Proteinase K
4. SDS 10%
5. Tris HCl
6. Phenol chloroform isoamyl alcohol (PCI)
7. Ethanol / isopropanol
8. TNE buffer

Reagents for DNA extraction (inorganic method)

1. Cell lysis buffer
2. NaCl/LiCl
3. TE buffer
4. Proteinase K
5. Ethanol

Reagents for RNA Purification from Blood

1. Trizol reagent
2. Isopropanol
3. 80% Ethanol
4. RNAse free water

Reagents for cDNA Synthesis

1. RNA template
2. 5X First Strand Buffer
3. 10mM dNTP Set
4. 0.1M DTT
5. Random Primers
6. RNaseOUT Ribonuclease Inhibitor
7. SuperScript II RNase H- Reverse Transcriptase
8. Water

FaceBook Whatsapp Instagram Youtube @Biotechbrainybunch

Reagents for PCR

Tris-HCl

- MgCl2
- Taq DNA polymerase
- dNTPs

• Template DNA (genomic, plasmid, cosmid, bacterial/yeast colony, etc.)

• Primers

• Water

Thermal cycler (MJ Research)

Reagents for Karyotyping

1. Alcohol,
2. methanol,
3. Glacial acetic acid, RPMI-1640, Streptomycin,
4. penicillin,
5. L-glutamine,
6. EtBr,
7. colchicine or colcemid, KCl,
8. Giemsa,
9. trypsin
10. water

FaceBook
Whatsapp
Instagram
Youtube
@Biotechbrainybunch

Reagents for Quantification of Nucleic Acids by Gel electrophoresis

1. Agaros gel
2. 1X TAE (Running buffer)
3. Ethidium bromide
4. Bromophenol blue (gel loading Dye)

Email; mehwishshabirhussain07@gmail.com |

REMEMBER IN YOUR PRAYERS

FaceBook
Whatsapp
Instagram
Youtube

@Biotechbrainybunch

Email; mehwishshabirhussain07@gmail.com |